Final Project Control System Design

Derivation 1

$$\xi = \begin{bmatrix} x & y & z \end{bmatrix}^T \tag{15.1}$$

$$\eta = \begin{bmatrix} \phi & \theta & \psi \end{bmatrix}^T \tag{15.2}$$

$$\nu = \begin{bmatrix} p & q & r \end{bmatrix}^T$$

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$R_z(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R(\phi,\theta,\psi) = R_z(\psi)R_y(\theta)R_x(\phi)$$

$$\nu = W\dot{\eta}$$

$$W = \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi\cos\theta \\ 0 & -\sin\phi & \cos\phi\cos\theta \end{bmatrix}$$

$$\dot{\eta} = W^{-1}\nu = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta \end{bmatrix}\nu \tag{15.3}$$

$$v_b = \begin{bmatrix} u & v & w \end{bmatrix}^T$$

$$\dot{\xi} = R(\phi,\theta,\psi)v_b \tag{15.5}$$

$$F_g = \begin{bmatrix} 0 & 0 & -mg \end{bmatrix}^T$$

$$F_{\text{thrust}} = \begin{bmatrix} 0 & 0 & u_T \end{bmatrix}^T$$

$$R(\phi,\theta,\psi)F_{\text{thrust}} \tag{15.6}$$

$$F_D = -k_D\dot{\xi} \tag{15.7}$$

$$f_i = k\omega_i^2, \quad i = 1,2,3,4,$$

$$u_T = \sum_i^4 f_i$$

$$\tau_B = \begin{bmatrix} u_\phi & u_\theta & u_\psi \end{bmatrix}^T = \begin{bmatrix} Lk(\omega_2^2 - \omega_4^2) \\ Lk(\omega_3^2 - \omega_1^2) \\ b(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2) \end{bmatrix} = \begin{bmatrix} 0 & L & 0 & -L \\ -L & 0 & L & 0 \\ \frac{b}{k} & -\frac{b}{k} & \frac{b}{k} & -\frac{b}{k} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}$$

$$I_n\dot{\nu} + v \times (I_n v) = \tau_B, \quad I_n = \text{diag}\{I_{xx}, I_{yy}, I_{zz}\} \tag{15.11}$$

$$\dot{\nu} = I_n^{-1}\tau_B + I_n^{-1}(\nu \times I_n\nu) \tag{15.12}$$

$$\ell\ddot{\alpha} = -\ddot{x}\cos\alpha - \ddot{z}\sin\alpha + g\sin\alpha \tag{15.16}$$

$$F_p = m_p(\ell\dot{\alpha}^2 - \ddot{x}\sin\alpha - \ddot{z}\cos\alpha - g\cos\alpha)\begin{bmatrix}\sin\alpha\\0\\\cos\alpha\end{bmatrix} \tag{15,18}$$

$$m\ddot{\xi} = R(\phi,\theta,\psi)\begin{bmatrix}0\\0\\u_T\end{bmatrix} + F_D + F_p + F_g$$

$$m\ddot{\xi} = R(\phi,\theta,\psi)\begin{bmatrix}0\\0\\u_T\end{bmatrix} - k_D\dot{\xi} + m_p(\ell\dot{\alpha}^2 - \ddot{x}\sin\alpha - \ddot{z}\cos\alpha - g\cos\alpha)\begin{bmatrix}\sin\alpha\\0\\\cos\alpha\end{bmatrix} + \begin{bmatrix}0\\0\\-mg\end{bmatrix}$$

$$m\ddot{\xi} = R(\phi,\theta,\psi)\begin{bmatrix}0\\0\\u_T\end{bmatrix} - k_D\dot{\xi} + m_p(\ell\dot{\alpha}^2 - g\cos\alpha)\begin{bmatrix}\sin\alpha\\0\\\cos\alpha\end{bmatrix} - m_p\begin{bmatrix}\sin^2\alpha & 0 & \cos\alpha\sin\alpha\\0 & 0 & 0\\\cos\alpha\sin\alpha & 0 & \cos^2\alpha\end{bmatrix}\ddot{\xi} + \begin{bmatrix}0\\0\\-mg\end{bmatrix}$$

$$(mI + m_p\begin{bmatrix}\sin^2\alpha & 0 & \cos\alpha\sin\alpha\\0 & 0 & 0\\\cos\alpha\sin\alpha & 0 & \cos^2\alpha\end{bmatrix})\ddot{\xi} = R(\phi,\theta,\psi)\begin{bmatrix}0\\0\\u_T\end{bmatrix} - k_D\dot{\xi} + m_p(\ell\dot{\alpha}^2 - g\cos\alpha)\begin{bmatrix}\sin\alpha\\0\\\cos\alpha\end{bmatrix} + \begin{bmatrix}0\\0\\-mg\end{bmatrix}$$

$$\begin{bmatrix}m + m_p\sin^2\alpha & 0 & m_p\cos\alpha\sin\alpha\\0 & m & 0\\m_p\cos\alpha\sin\alpha & 0 & m + m_p\cos^2\alpha\end{bmatrix}\ddot{\xi} = R(\phi,\theta,\psi)\begin{bmatrix}0\\0\\u_T\end{bmatrix} - k_D\dot{\xi} + m_p(\ell\dot{\alpha}^2 - g\cos\alpha)\begin{bmatrix}\sin\alpha\\0\\\cos\alpha\end{bmatrix} + \begin{bmatrix}0\\0\\-mg\end{bmatrix}$$

$$\ddot{\xi} = \frac{1}{m(m+m_p)}\begin{bmatrix}m + m_p\cos^2\alpha & 0 & -m_p\sin\alpha\cos\alpha\\0 & m + m_p & 0\\-m_p\sin\alpha\cos\alpha & 0 & m + m_p\sin^2\alpha\end{bmatrix}\left(R(\phi,\theta,\psi)\begin{bmatrix}0\\0\\u_T\end{bmatrix} - k_D\dot{\xi} + m_p(l\dot{\alpha}^2 - g\cos\alpha)\begin{bmatrix}\sin\alpha\\0\\\cos\alpha\end{bmatrix} + \begin{bmatrix}0\\0\\-mg\end{bmatrix}\right)$$

Taking just what is inside the parenthesis:

$$R\begin{bmatrix}0\\0\\u_T\end{bmatrix} - k_D\dot{\xi} + m_p l\dot{\alpha}^2 - g\begin{bmatrix}m_p\sin\alpha\cos\alpha\\0\\m + m_p\cos^2\alpha\end{bmatrix}$$

$$\ddot{\alpha} = -\frac{1}{\ell}[\cos\alpha \quad 0 \quad \sin\alpha]\,\ddot{\xi} + \frac{g}{\ell}\sin\alpha$$

$$\ddot{\alpha} = \frac{1}{m + m_p}([-\cos\alpha \quad 0 \quad \sin\alpha]\,(R\begin{bmatrix}0\\0\\u_T\end{bmatrix} - k_D\dot{\xi}))$$

Let ξ, $\dot{\xi}$, η, ν, α, $\dot{\alpha}$ be our states, then

$$\frac{\partial}{\partial t}\begin{bmatrix}\xi\\\dot{\xi}\\\eta\\\nu\\\alpha\\\dot{\alpha}\end{bmatrix} = \begin{bmatrix}\dot{\xi}\\\frac{1}{m(m+m_p)}\begin{bmatrix}m + m_p\cos^2\alpha & 0 & -m_p\sin\alpha\cos\alpha\\0 & m + m_p & 0\\-m_p\sin\alpha\cos\alpha & 0 & m + m_p\sin^2\alpha\end{bmatrix}\left(R(\phi,\theta,\psi)\begin{bmatrix}0&0&0&0\\0&0&0&0\\1&1&1&1\end{bmatrix}\begin{bmatrix}f_1\\f_2\\f_3\\f_4\end{bmatrix} - k_D\dot{\xi} + m_p l\dot{\alpha}^2 - g\begin{bmatrix}m_p\sin\alpha\cos\\0\\m + m_p\cos\end{bmatrix}\right.\\W^{-1}(\phi,\theta)\nu\\I_n^{-1}\begin{bmatrix}0 & L & 0 & -L\\-L & 0 & L & 0\\\frac{b}{k} & -\frac{b}{k} & \frac{b}{k} & -\frac{b}{k}\end{bmatrix}\begin{bmatrix}f_1\\f_2\\f_3\\f_4\end{bmatrix} + I_n^{-1}(\nu\times I_n\nu)\\\dot{\alpha}\\\frac{1}{m+m_p}([-\cos\alpha \quad 0 \quad \sin\alpha]\,(R(\phi,\theta,\psi)\begin{bmatrix}0&0&0&0\\0&0&0&0\\1&1&1&1\end{bmatrix}\begin{bmatrix}f_1\\f_2\\f_3\\f_4\end{bmatrix} - k_D\dot{\xi}))\end{bmatrix}$$

$$\begin{bmatrix} u_T \\ \tau_B \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & L & 0 & -L \\ -L & 0 & L & 0 \\ \frac{b}{k} & -\frac{b}{k} & \frac{b}{k} & -\frac{b}{k} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}$$

$$\frac{\partial}{\partial t}\begin{bmatrix} \xi \\ \dot{\xi} \\ \eta \\ \nu \\ \alpha \\ \dot{\alpha} \end{bmatrix} = \begin{bmatrix} \dot{\xi} \\ \frac{1}{m(m+m_p)} \begin{bmatrix} m+m_p\cos^2\alpha & 0 & -m_p\sin\alpha\cos\alpha \\ 0 & m+m_p & 0 \\ -m_p\sin\alpha\cos\alpha & 0 & m+m_p\sin^2\alpha \end{bmatrix} \left(R(\phi,\theta,\psi)\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} u_T \\ \tau_B \end{bmatrix} - k_D\dot{\xi} + m_p l\dot{\alpha}^2 - g\begin{bmatrix} m_p\sin\alpha\cos\alpha \\ 0 \\ m+m_p\cos^2\alpha \end{bmatrix} \right) \\ W^{-1}(\phi,\theta)\nu \\ I_n^{-1}\begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix}\begin{bmatrix} u_T \\ \tau_B \end{bmatrix} + I_n^{-1}(\nu \times I_n\nu) \\ \dot{\alpha} \\ \frac{1}{m+m_p}([-\cos\alpha \quad 0 \quad \sin\alpha]\,(R(\phi,\theta,\psi)\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} u_T \\ \tau_B \end{bmatrix} - k_D\dot{\xi})) \end{bmatrix}$$

$$M_g(\alpha) = \begin{bmatrix} m_p\sin\alpha\cos\alpha \\ 0 \\ m+m_p\cos^2\alpha \end{bmatrix}, A_t^{-1}(\alpha) = \begin{bmatrix} -\frac{\cos\alpha}{m+m_p} & 0 & \frac{\sin\alpha}{m+m_p} \end{bmatrix}, P_t^{-1}(\alpha) = \begin{bmatrix} \frac{m+m_p\cos^2\alpha}{m(m+m_p)} & 0 & -\frac{m_p\sin\alpha\cos\alpha}{m(m+m_p)} \\ 0 & \frac{1}{m} & 0 \\ -\frac{m_p\sin\alpha\cos\alpha}{m(m+m_p)} & 0 & \frac{m+m_p\sin^2\alpha}{m(m+m_p)} \end{bmatrix}$$

$$\frac{\partial}{\partial t}\begin{bmatrix} \xi \\ \dot{\xi} \\ \eta \\ \nu \\ \alpha \\ \dot{\alpha} \end{bmatrix} = \begin{bmatrix} \dot{\xi} \\ P_t^{-1}(\alpha)\left(R(\phi,\theta,\psi)\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} u_T \\ \tau_B \end{bmatrix} - k_D\dot{\xi} + m_p l\dot{\alpha}^2 - gM_g(\alpha) \right) \\ W^{-1}(\phi,\theta)\nu \\ I_n^{-1}\begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix}\begin{bmatrix} u_T \\ \tau_B \end{bmatrix} + I_n^{-1}(\nu \times I_n\nu) \\ \dot{\alpha} \\ A_t^{-1}(\alpha)\left(R(\phi,\theta,\psi)\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} u_T \\ \tau_B \end{bmatrix} - k_D\dot{\xi} \right) \end{bmatrix}$$

Default Nonlinear System

Therefore the autonomous portion is:

$$\frac{\partial}{\partial t}\begin{bmatrix} \xi \\ \dot{\xi} \\ \eta \\ \nu \\ \alpha \\ \dot{\alpha} \end{bmatrix} = \begin{bmatrix} \dot{\xi} \\ P_t^{-1}(\alpha)\left(-k_D\dot{\xi} + m_p\ell\dot{\alpha}^2 - gM_g(\alpha) \right) \\ W^{-1}(\eta)\nu \\ I_n^{-1}[\nu]_\times I_n\nu \\ \dot{\alpha} \\ -k_D A_t^{-1}(\alpha)\dot{\xi} \end{bmatrix}$$

Where $[\nu]_\times$ is the skew-symmetric matrix of ν.
And the input portion is (B matrix):

$$\frac{\partial}{\partial t}\begin{bmatrix} \xi \\ \dot{\xi} \\ \eta \\ \nu \\ \alpha \\ \dot{\alpha} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ P_t^{-1}(\alpha)R(\eta)\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} & 0 \\ 0 & 0 \\ 0 & I_n^{-1} \\ 0 & 0 \\ A_t^{-1}(\alpha)R(\eta)\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} & 0 \end{bmatrix}\begin{bmatrix} u_T \\ \tau_B \end{bmatrix}$$

If we define R_3 to be the third column of the rotation matrix:

$$\frac{\partial}{\partial t}\begin{bmatrix}\xi\\ \dot{\xi}\\ \eta\\ \nu\\ \alpha\\ \dot{\alpha}\end{bmatrix} = \begin{bmatrix}0 & 0\\ P_t^{-1}(\alpha)R_3(\eta) & 0\\ 0 & 0\\ 0 & I_n^{-1}\\ 0 & 0\\ A_t^{-1}(\alpha)R_3(\eta) & 0\end{bmatrix}\begin{bmatrix}u_T\\ \tau_B\end{bmatrix}$$

The Jacobian of the autonomous portion is (A matrix):

$$\begin{bmatrix}0 & I & 0 & 0 & 0 & 0\\ 0 & -k_D P_t^{-1} & 0 & 0 & P_{t,a}^{-1}\left(-k_D\dot{\xi} + m_p\ell\dot{\alpha}^2 - gM_g\right) - gP_t^{-1}M_{g,a} & 2m_p\ell P_t^{-1}\dot{\alpha}\\ 0 & 0 & W_\eta^{-1}\nu & W^{-1} & 0 & 0\\ 0 & 0 & 0 & I_n^{-1}([\nu]_\times I_n - [I_n\nu]_\times) & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 1\\ 0 & -k_D A_t^{-1} & 0 & 0 & -k_D A_{t,\alpha}^{-1}\dot{\xi} & 0\end{bmatrix}$$

Note subscript is derivative, so $P_{t,\alpha}^{-1} = \frac{\partial}{\partial\alpha}P_t^{-1}(\alpha)$. Vector derivatives (like W_η^{-1}) are tensors, so each case is its own matrix multiplication.

Important matrices

$$M_g(\alpha) = \begin{bmatrix}m_p\sin\alpha\cos\alpha\\ 0\\ m + m_p\cos^2\alpha\end{bmatrix}$$

$$A_t^{-1}(\alpha) = \begin{bmatrix}-\frac{\cos\alpha}{m+m_p} & 0 & \frac{\sin\alpha}{m+m_p}\end{bmatrix}$$

$$P_t^{-1}(\alpha) = \begin{bmatrix}\frac{m+m_p\cos^2\alpha}{m(m+m_p)} & 0 & -\frac{m_p\sin\alpha\cos\alpha}{m(m+m_p)}\\ 0 & \frac{1}{m} & 0\\ -\frac{m_p\sin\alpha\cos\alpha}{m(m+m_p)} & 0 & \frac{m+m_p\sin^2\alpha}{m(m+m_p)}\end{bmatrix}$$

$$R_3(\eta) = \begin{bmatrix}\cos\psi\sin\theta\cos\phi + \sin\psi\sin\phi\\ \sin\psi\sin\theta\cos\phi - \cos\psi\sin\phi\\ \cos\theta\cos\phi\end{bmatrix}$$

$$W^{-1}(\eta) = \begin{bmatrix}1 & \sin\phi\tan\theta & \cos\phi\tan\theta\\ 0 & \cos\phi & -\sin\phi\\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta\end{bmatrix}$$

Constants

$$I_n = \begin{bmatrix}I_{xx} & 0 & 0\\ 0 & I_{yy} & 0\\ 0 & 0 & I_{zz}\end{bmatrix}, \ell, k_D, m, m_p, g$$

Note

$$\begin{bmatrix}u_T\\ \tau_B\end{bmatrix} = \begin{bmatrix}1 & 1 & 1 & 1\\ 0 & L & 0 & -L\\ -L & 0 & L & 0\\ \frac{b}{k} & -\frac{b}{k} & \frac{b}{k} & -\frac{b}{k}\end{bmatrix}\begin{bmatrix}f_1\\ f_2\\ f_3\\ f_4\end{bmatrix}$$

Which relates our control inputs to the motor forces, where the maximum force constraint is applied. This matrix is invertible.

Derivation 2

Now first say that we want to change a state $\alpha \to s_\alpha = \sin(\alpha)$

$$\dot{s}_\alpha = \dot{\alpha}\cos(\alpha) = \dot{\alpha}\sqrt{1-s_\alpha^2}$$

$$\frac{\partial}{\partial t}\begin{bmatrix}\xi\\ \dot{\xi}\\ \eta\\ \nu\\ s_\alpha\\ \dot{\alpha}\end{bmatrix} = \begin{bmatrix}\dot{\xi}\\ P_t^{-1}(s_\alpha)\left(-k_D\dot{\xi} + m_p l\dot{\alpha}^2 - gM_g(s_\alpha)\right)\\ W^{-1}(\eta)\nu\\ I_n^{-1}[\nu]_\times I_n\nu\\ \dot{\alpha}\sqrt{1-s_\alpha^2}\\ -k_D A_t^{-1}(s_\alpha)\dot{\xi}\end{bmatrix}$$

$$P_t^{-1}(s_\alpha) = \begin{bmatrix}\frac{m+m_p(1-s_\alpha^2)}{m(m+m_p)} & 0 & -\frac{m_p s_\alpha\sqrt{1-s_\alpha^2}}{m(m+m_p)}\\ 0 & \frac{1}{m} & 0\\ -\frac{m_p s_\alpha\sqrt{1-s_\alpha^2}}{m(m+m_p)} & 0 & \frac{m+m_p s_\alpha^2}{m(m+m_p)}\end{bmatrix}$$

$$M_g(s_\alpha) = \begin{bmatrix}m_p s_\alpha\sqrt{1-s_\alpha^2}\\ 0\\ m+m_p(1-s_\alpha^2)\end{bmatrix}$$

$$A_t^{-1}(s_\alpha) = \begin{bmatrix}-\frac{\sqrt{1-s_\alpha^2}}{m+m_p} & 0 & \frac{s_\alpha}{m+m_p}\end{bmatrix}$$

$$\begin{bmatrix}\dot{\phi}\\ \theta\\ \psi\end{bmatrix} = \begin{bmatrix}1 & \sin\phi\tan\theta & \cos\phi\tan\theta\\ 0 & \cos\phi & -\sin\phi\\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta\end{bmatrix}\nu$$

$$s_\phi = \sin(\phi)$$

$$\dot{s_\phi} = \dot{\phi}\cos(\phi)$$

$$\begin{bmatrix}\dot{s_\phi}\\ s_\theta\\ s_\psi\end{bmatrix} = \begin{bmatrix}\cos\phi & \sin\phi\cos\phi\frac{\sin\theta}{\cos\theta} & \cos^2\phi\frac{\sin\theta}{\cos\theta}\\ 0 & \cos\phi\cos\theta & -\sin\phi\cos\theta\\ 0 & \frac{\sin\phi\cos\psi}{\cos\theta} & \frac{\cos\phi\cos\psi}{\cos\theta}\end{bmatrix}v$$

$$\begin{bmatrix}\dot{s_\phi}\\ s_\theta\\ s_\psi\end{bmatrix} = \begin{bmatrix}\sqrt{1-s_\phi^2} & s_\phi s_\theta\sqrt{\frac{1-s_\phi^2}{1-s_\theta^2}} & \sqrt{1-s_\phi^2}s_\theta\sqrt{\frac{1-s_\phi^2}{1-s_\theta^2}}\\ 0 & \sqrt{(1-s_\phi^2)(1-s_\theta^2)} & -s_\phi\sqrt{1-s_\theta^2}\\ 0 & s_\phi\sqrt{\frac{1-s_\psi^2}{1-s_\theta^2}} & \sqrt{1-s_\phi^2}\sqrt{\frac{1-s_\psi^2}{1-s_\theta^2}}\end{bmatrix}v$$

$$W^{-1}(s_\eta) = \begin{bmatrix}\sqrt{1-s_\phi^2} & s_\phi s_\theta\sqrt{\frac{1-s_\phi^2}{1-s_\theta^2}} & \sqrt{1-s_\phi^2}s_\theta\sqrt{\frac{1-s_\phi^2}{1-s_\theta^2}}\\ 0 & \sqrt{(1-s_\phi^2)(1-s_\theta^2)} & -s_\phi\sqrt{1-s_\theta^2}\\ 0 & s_\phi\sqrt{\frac{1-s_\psi^2}{1-s_\theta^2}} & \sqrt{1-s_\phi^2}\sqrt{\frac{1-s_\psi^2}{1-s_\theta^2}}\end{bmatrix}$$

$$R_x(s_\phi) = \begin{bmatrix}1 & 0 & 0\\ 0 & \sqrt{1-s_\phi^2} & -s_\phi\\ 0 & s_\phi & \sqrt{1-s_\phi^2}\end{bmatrix}$$

$$R_y(s_\theta) = \begin{bmatrix}\sqrt{1-s_\theta^2} & 0 & s_\theta\\ 0 & 1 & 0\\ -s_\theta & 0 & \sqrt{1-s_\theta^2}\end{bmatrix}$$

$$R_z(s_\psi) = \begin{bmatrix} \sqrt{1-s_\psi^2} & -s_\psi & 0 \\ s_\psi & \sqrt{1-s_\psi^2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_3 = R_z(s_\psi) R_y(s_\theta) R_x(s_\phi) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$R_3 = R_z(s_\psi) R_y(s_\theta) \begin{bmatrix} 0 \\ -s_\phi \\ \sqrt{1-s_\phi^2} \end{bmatrix}$$

$$R_3 = R_z(s_\psi) \begin{bmatrix} s_\theta \sqrt{1-s_\phi^2} \\ -s_\phi \\ \sqrt{1-s_\theta^2}\sqrt{1-s_\phi^2} \end{bmatrix}$$

$$R_3(s_\eta) = \begin{bmatrix} s_\theta \sqrt{1-s_\phi^2}\sqrt{1-s_\psi^2} + s_\phi s_\psi \\ s_\psi s_\theta \sqrt{1-s_\phi^2} - s_\phi \sqrt{1-s_\psi^2} \\ \sqrt{1-s_\theta^2}\sqrt{1-s_\phi^2} \end{bmatrix}$$

Therefore:

Sine Nonlinear System

$$\frac{\partial}{\partial t} \begin{bmatrix} \xi \\ \dot{\xi} \\ s_\eta \\ \nu \\ s_\alpha \\ \dot{\alpha} \end{bmatrix} = \begin{bmatrix} \dot{\xi} \\ P_t^{-1}(s_\alpha)\left(-k_D\dot{\xi} + m_p \ell \dot{\alpha}^2 - g M_g(s_\alpha)\right) \\ W^{-1}(s_\eta)\nu \\ I_n^{-1}[\nu]_\times I_n \nu \\ \dot{\alpha}\sqrt{1-s_\alpha^2} \\ -k_D A_t^{-1}(s_\alpha)\dot{\xi} \end{bmatrix}$$

and

$$\frac{\partial}{\partial t} \begin{bmatrix} \xi \\ \dot{\xi} \\ s_\eta \\ \nu \\ s_\alpha \\ \dot{\alpha} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ P_t^{-1}(s_\alpha) R_3(s_\eta) & 0 \\ 0 & 0 \\ 0 & I_n^{-1} \\ 0 & 0 \\ A_t^{-1}(s_\alpha) R_3(s_\eta) & 0 \end{bmatrix} \begin{bmatrix} u_T \\ \tau_B \end{bmatrix}$$

With Jacobian:

$$\begin{bmatrix} 0 & I & 0 & 0 & 0 & 0 \\ 0 & -k_D P_t^{-1} & 0 & 0 & P_{t,s_\alpha}^{-1}\left(-k_D\dot{\xi} + m_p\ell\dot{\alpha}^2 - gM_g\right) - gP_t^{-1}M_{g,s_\alpha} & 2m_p\ell P_t^{-1}\dot{\alpha} \\ 0 & 0 & W_{s_\eta}^{-1}\nu & W^{-1} & 0 & 0 \\ 0 & 0 & 0 & I_n^{-1}([\nu]_\times I_n - [I_n\nu]_\times) & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\dot{\alpha}s_\alpha}{\sqrt{1-s_\alpha^2}} & \sqrt{1-s_\alpha^2} \\ 0 & -k_D A_t^{-1} & 0 & 0 & -k_D A_{t,s_\alpha}^{-1}\dot{\xi} & 0 \end{bmatrix}$$

Important matrices

$$M_g(s_\alpha) = \begin{bmatrix} m_p s_\alpha \sqrt{1-s_\alpha^2} \\ 0 \\ m + m_p(1-s_\alpha^2) \end{bmatrix}$$

$$A_t^{-1}(s_\alpha) = \begin{bmatrix} -\frac{\sqrt{1-s_\alpha^2}}{m+m_p} & 0 & \frac{s_\alpha}{m+m_p} \end{bmatrix}$$

$$P_t^{-1}(s_\alpha) = \begin{bmatrix} \frac{m+m_p(1-s_\alpha^2)}{m(m+m_p)} & 0 & -\frac{m_p s_\alpha \sqrt{1-s_\alpha^2}}{m(m+m_p)} \\ 0 & \frac{1}{m} & 0 \\ -\frac{m_p s_\alpha \sqrt{1-s_\alpha^2}}{m(m+m_p)} & 0 & \frac{m+m_p s_\alpha^2}{m(m+m_p)} \end{bmatrix}$$

$$R_3(s_\eta) = \begin{bmatrix} s_\theta \sqrt{1-s_\phi^2}\sqrt{1-s_\psi^2} + s_\phi s_\psi \\ s_\psi s_\theta \sqrt{1-s_\phi^2} - s_\phi \sqrt{1-s_\psi^2} \\ \sqrt{1-s_\theta^2}\sqrt{1-s_\phi^2} \end{bmatrix}$$

$$W^{-1}(s_\eta) = \begin{bmatrix} \sqrt{1-s_\phi^2} & s_\phi s_\theta \sqrt{\frac{1-s_\phi^2}{1-s_\theta^2}} & \sqrt{1-s_\phi^2} s_\theta \sqrt{\frac{1-s_\phi^2}{1-s_\theta^2}} \\ 0 & \sqrt{(1-s_\phi^2)(1-s_\theta^2)} & -s_\phi \sqrt{1-s_\theta^2} \\ 0 & s_\phi \sqrt{\frac{1-s_\psi^2}{1-s_\theta^2}} & \sqrt{1-s_\phi^2}\sqrt{\frac{1-s_\psi^2}{1-s_\theta^2}} \end{bmatrix}$$

Note $W^{-1}(s_\eta)$ is actually $\begin{bmatrix} \sqrt{1-s_\phi^2} & 0 & 0 \\ 0 & \sqrt{1-s_\theta^2} & 0 \\ 0 & 0 & \sqrt{1-s_\psi^2} \end{bmatrix}$ times the direct substitution of $W^{-1}(\alpha)$, different from all other updated matrices, where we directly substituted $\sin\alpha \to s_\alpha$,$\cos\alpha \to \sqrt{1-s_\alpha^2}$ or similar.

Constants

$$I_n = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}, \ell, k_D, m, m_p, g$$

Justification

First, these systems are only equal under the assumption that $\alpha, \eta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, however linearization fails before this assumption. Approximately we can say there are 5 approximated nonlinear functions of interest: $\sin x$, $\cos x$, $\sin^2 x$, $\cos^2 x$, and $\sin x \cos x$. All matrices above are composed from these 5 functions were inspecting one variable. Say that we linearize with respect to x:

$$\begin{aligned} \sin x &\to x \\ \cos x &\to 1 \\ \sin^2 x &\to 0 \\ \cos^2 x &\to 1 \\ \sin x \cos x &\to x \end{aligned}$$

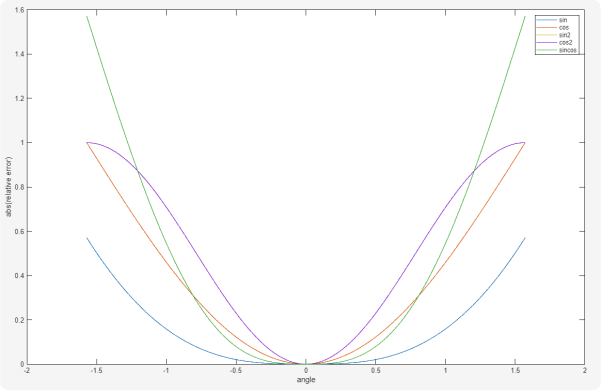


Figure 1: absolute value of absolute error between nonlinear and linear wrt x

Note that the highest error comes from the linearization of $\sin x \cos x$.
If instead we linearize with respect to $\sin(x)$, then we have the following transformations:

$$\begin{aligned} \sin x &\to \sin x \\ \cos x &\to 1 \\ \sin^2 x &\to 0 \\ \cos^2 x &\to 1 \\ \sin x \cos x &\to \sin x \end{aligned}$$

The only two that change is $\sin x$ and $\sin x \cos x$, and only $\sin x \cos x$ has error:

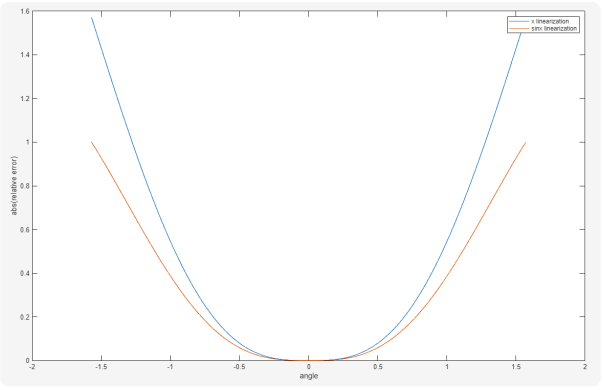


Figure 2: comparison of error of sin x cos x linearization between x and sin x

Plotting all errors:



Figure 3: absolute value of absolute error between nonlinear and linear wrt sin x

So the maximum error over the region has been decreased from $\frac{\pi}{2}$ to 1 with the change of variables.